



**Project Title: "Student Performance & Attendance Tracker"**

## Objective

Develop a **Student Performance & Attendance Tracker** using **MySQL**, where institutions can manage student records, monitor attendance, track performance across subjects, and generate reports. The system should support CRUD operations, filtering, sorting, aggregation, key relationships, joins, subqueries, and advanced SQL functions.

### Problem Statement

**Design a Student Performance & Attendance Tracker** that enables educational institutions to:

1. **Manage Student & Faculty Information** (add, update, delete, search).
2. **Track Student Attendance** (daily logging, percentage calculation, and absence reports).
3. **Store and Analyze Student Performance** (marks, grades, subject-wise trends).
4. **Generate Insights & Reports** (top-performing students, attendance defaulters).

## Database Schema

## Tables & Relationships

- ## 1. Students

- o student\_id (Primary Key)
- o name
- o dob
- o gender
- o email
- o phone\_number







- Retrieve students who have attendance **below 50% AND** are failing.
- Find students who scored **above 90 OR** have perfect attendance.
- List faculty members **NOT** assigned to any course.

#### 4. Sorting & Grouping Data (ORDER BY, GROUP BY) (Medium Weightage)

- List students **alphabetically by name**.
- Count the number of students enrolled in each department.
- Show the **average marks per course**.

### 5. Use Aggregate Functions (SUM, AVG, MAX, MIN, COUNT) (High Weightage)

- Find the **average attendance percentage** of students.
- Identify the **highest and lowest marks obtained** in each course.
- Calculate the **total number of students per department**.

### 6. Establish Primary & Foreign Key Relationships (High Weightage)

- Ensure that students cannot enroll in the same course multiple times.
- Link faculty members to the courses they teach.

## 7. Implement Joins (High Weightage)

- Retrieve student details along with their department using **INNER JOIN**.
- Get a list of students who have not enrolled in any course using **LEFT JOIN**.

"Quality is our Motto."

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*Shaping "skills" for "scaling" higher...!!!*

- List courses that have no faculty assigned using **RIGHT JOIN**.
- Show students **without** grades using **FULL OUTER JOIN**.

### 8. Use Subqueries (High Weightage)





